

Townsend 5.7-5.8

Questions for Gallicchio

What is the reduced density operator?

5.7

The density operator tells us the particle in a region of phase space instead of a pure single state - since we can't perfectly separate out particles in real experiment.

For a pure state

$$\hat{\rho} = |\psi\rangle\langle\psi|$$

which is the identity. Non pure mixed states have $\hat{\rho}^2 \leq 1$.

Multiparticle systems

$$\hat{\rho} = |\psi_{12}\rangle\langle\psi_{12}| = (|+\mathbf{z}\rangle_1|+\mathbf{z}\rangle_2)(\langle+\mathbf{z}|_1\langle+\mathbf{z}|_2)$$

the reduced density operator?

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